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Quantifying stability of parameter estimates for *in vivo* nearly incompressible transversely-isotropic brain MR elastography

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Abstract

Easily computable quality metrics for measured medical data at point-of-care are important for imaging technologies involving offline reconstruction. Accordingly, we developed a new data quality metric for *in vivo* transversely-isotropic (TI) magnetic resonance elastography (MRE) based on a generalization of the widely accepted octahedral shear-strain calculation. The metric uses MRE displacement data and an estimate of the TI property field to yield a 'stability map' which predicts regions of low versus high accuracy in the resulting material property reconstructions. We can also calculate an average TI parameter stability (TIPS) score over all voxels in a region of interest for a given measurement to indicate how reliable the recovered mechanical property estimate for the region is expected to be. The calculation is rapid and places little demand on computing resources compared to the computationally intensive material property reconstruction from non-linear inversion (TI-NLI) of displacement fields, making it ideal for point-of-care evaluation of data quality. We test the predictions of the stability map for both simulated phantoms and *in vivo* human brain data. We used a range of different displacement datasets from vibrations applied in the anterior-posterior (AP), left-right (LR) and combined AP + LR directions. The TIPS and variability maps (noise sensitivity or variation from the mean of repeated MRE scans) were consistently anti-correlated. Notably, Spearman correlation coefficients |R|>0.6 were found between variability and TIPS score for individual white matter tracts with *in vivo* data. These observations demonstrate the reliability and promise of this data quality metric to screen data rapidly in realistic clinical MRE applications.

Keywords

MR elastography; anisotropy; brain; *in vivo* ; data quality metric; octahedral shear strain; material property reconstruction

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Introduction

Magnetic resonance elastography (MRE) generates images of the mechanical properties of tissue using displacement fields measured with phase contrast MRI. Typically, an external actuator is used to vibrate the body surface, and the resulting mechanical waves propagate into the tissue to probe the mechanical properties. MRE has had considerable clinical success in detecting liver fibrosis, and numerous recent studies have used MRE in the brain to diagnose and monitor diseases including Alzheimer's disease [1–3], multiple sclerosis [4, 5], intracranial tumors [6, 7], epilepsy [8], and amyotrophic lateral sclerosis [9]. MRE in healthy brain has also uncovered structure-function relationships where the measured mechanical properties of brain structures are predictive of functions such as memory and fluid intelligence [10–12]. The reliability and usefulness of MRE metrics in characterizing brain health or function depends on accuracy and precision of the property estimates. A number of factors affect the results of an MRE exam, and data quality metrics are valuable to ensure data quality of a given exam is sufficient for a given application.

Some MRE tissues of interest consist of tracts of aligned fibers, such as muscle and brain white matter, which are likely to produce an anisotropic mechanical response. This reality has motivated the development of anisotropic MRE methods [13–17] that can return direction-dependent mechanical properties of tissue. Here, predictive stability measures are important due to the directional dependence of the stress-strain relationship; some wave fields generate stresses completely independent of specific anisotropic stiffness parameters, thus making their estimation inaccurate or even impossible [18]. In these particular cases, or particular regions within a larger data set, a model-based inverse method may be unable to estimate some mechanical properties accurately, and a data quality metric that describes the expected stability of parameter estimates would provide insight into the reliability of outputs.

In this paper, we present a generalization of the octahedral shear strain-based signal-to-noise (OSS-SNR) measurement [19] for the case of a nearly incompressible transversely isotropic material. We demonstrate that the stability measure is predictive of inversion stability for simulated data where the ground truth is known, and that it can identify in advance which of a series of 10 repeated *in vivo* scans will deviate the most from mean values. The approach presented here can be generalized for nearly any candidate material model.

Methods

Nearly incompressible transversely isotropic model

The time-harmonic nearly incompressible transversely isotropic model used in our transversely isotropic MRE inversion algorithm is described in a Cartesian coordinate system by the 6-dimensional Mandel–Voigt representation [20]

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$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{Bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{21} & c_{22} & c_{23} & 0 & 0 & 0 \\ c_{31} & c_{32} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{Bmatrix} \tag{1}$$

where σ_{ij} and ε_{ij} are complex-valued time-harmonic stress and strain amplitudes of the form $\sigma(x,t) = Re\left(\sigma(x)e^{i\omega t}\right)$. Defining the fiber axis as the x_1 direction, the components of the 6 × 6 elasticity matrix, $[\boldsymbol{C}]$, are given by

$$\begin{aligned} c_{11} &= \kappa + \frac{4}{3}\mu\left(1 + \frac{4}{3}\zeta\right),\ c_{22} = c_{33} \\ &= \kappa + \frac{4}{3}\mu\left(1 + \frac{1}{3}\zeta\right),\ c_{44} = c_{66} = \mu(1+\phi), \\ c_{12} &= c_{13} = c_{21} = c_{31} = \kappa - \frac{2}{3}\mu\left(1 + \frac{4}{3}\zeta\right), \\ c_{32} &= c_{23} = \kappa - \frac{2}{3}\mu\left(1 - \frac{2}{3}\zeta\right),\ c_{55} = \mu \end{aligned} \tag{2}$$

μ is a complex-valued base shear modulus which imparts an isotropic form of damping where the damping ratio is $\xi = \frac{Im(\mu)}{2Re(\mu)}$. ϕ is the shear anisotropy such that the shear modulus in planes parallel to the fibers is $\mu(1+\phi)$, ζ is the tensile anisotropy such that the tensile modulus along the fibers is $E(1+\zeta)$, and κ is the isotropic bulk modulus. The Mandel–Voigt representation aligned with the fiber direction can be rotated into a global coordinate system using the appropriate 6 × 6 rotation matrix, B, such that [21, 22] $\{\sigma_{global}\} = [B]\{\sigma_{local}\}$.

The resulting stress tensor, and the definition of the strain tensor in terms of the displacement vector, u, $\varepsilon_{ij} = \left(\frac{du_i}{\partial x_j} + \frac{du_j}{\partial x_i}\right)$ is applied to the equation describing steady-state vibration of a viscoelastic solid, $\nabla \cdot \sigma = -\rho\omega^2 u$, to give the TI equations of motion, which are written in a displacement-pressure formulation on finite elements and are valid for nearly incompressible, heterogenous isotropic and transversely-isotropic media [22]. Throughout this paper, both for simulation and *in vivo* data, we use 50 Hz as the actuation frequency.

Transversely isotropic nonlinear inversion (TI-NLI)

The model-based MRE algorithm [17] used in this work defines the fiber direction based on the primary eigenvector from diffusion tensor imaging. The algorithm starts with an estimate of the mechanical property distribution, $\theta(\vec{x})$, and minimizes the difference between measured displacements, $\vec{u}_m$ (arrow indicating vector in Cartesian space), and displacements from a finite element computational model, $\vec{u}_c(\theta(\vec{x}))$, by iteratively updating the property direction using quasi-Newton gradient descent to minimize $\Phi = \sum(\vec{u}_c(\theta(\vec{x})) - \vec{u}_m)$. The large computational cost of numerous forward solutions of a

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potentially large 3D finite element system is mitigated by breaking the problem into a set of overlapping subzones which are processed in parallel on a distributed computing cluster. Boundary conditions for each of the subzones are taken from measured data, and multiple sets of measured data (e.g. from different boundary conditions, actuation directions, or vibration frequencies) are used to recover a single set of mechanical properties by expanding the summation over multiple measurement sets. Inversion parameters chosen for all of the experiments described in this paper were the same as those applied during *in vivo* TI-NLI to ensure clinical relevance.

Although adjusting the TI-NLI regularization parameters can improve contrast recovery in idealized simulated data, as in our previous MRE work we use the same inversion parameters for simulated and *in vivo* data to ensure conclusions drawn from simulations are applicable to clinical practice. The regularization required for stability with noisy *in vivo* data results in some loss of contrast recovery in smaller structures though it improves when more actuation directions are included.

Transversely isotropic parameter stability (TIPS) measure

We hypothesize that the stability of a particular NITI parameter is related to how sensitive the local stress field is to changes in that parameter value. If changes in a parameter have a relatively small effect on the stress at a given deformation state, the parameter estimate will have a greater change with perturbations in the MRE motion data due to noise and imaging artifacts, and thus, be considered less stable. For a given estimate of the NITI parameters, and measurements of the strain field, ε, the change in the stress field when parameter θ_i is perturbed at a location $\vec{x}$, is given by

$$\Delta_{\sigma,\theta_i}(\vec{x}) = \left\| \frac{\partial \sigma\left(\theta(\vec{x}), \varepsilon(\vec{x})\right)}{\partial \theta_i} \right\|_2 \tag{3}$$

where $\| \cdot \|_2$ is the matrix 2-norm for the stress tensor in Voigt notation. Larger values of this TIPS measure, $\Delta_{\sigma,\theta_i}(\vec{x})$, will provide more stable MRE property estimates at this location.

This formalism is a general case of the widely used octahedral shear strain (OSS) as a data quality measure in isotropic MRE—OSS is equal to the derivative of the maximum shear stress with respect to the isotropic shear modulus. The stress tensor in the global coordinate system for the NITI model is $\sigma(\theta, \varepsilon, \vec{x}) = [B_\epsilon]^T[C][B_\epsilon][\varepsilon]$, which gives the required sensitivity derivatives as $\frac{\partial \sigma(\theta, \varepsilon, \vec{x})}{\partial \theta_i} = [B_\epsilon]^T\left[\frac{\partial C}{\partial \theta_i}\right][B_\epsilon][\varepsilon]$. The components of $\left[\frac{\partial C}{\partial \theta_i}\right]$ are derived by differentiating equation (1), where the nonzero terms are

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$$\begin{aligned}
&\frac{\partial c_{11}}{\partial \mu} = \frac{4}{3}\left(1 + \frac{4}{3}\zeta\right), \frac{\partial c_{11}}{\partial \zeta} = \frac{16}{9}\mu; \\
&\frac{\partial c_{22|33}}{\partial \mu} = \frac{4}{3}\left(1 + \frac{1}{3}\zeta\right), \frac{\partial c_{22|33}}{\partial \zeta} = \frac{4}{9}\mu \\
&\frac{\partial c_{44|66}}{\partial \mu} = (1 + \phi), \frac{\partial c_{44|66}}{\partial \phi} = \mu, \\
&\frac{\partial c_{12|13|21|31}}{\partial \mu} = -\frac{2}{3}\left(1 + \frac{4}{3}\zeta\right), \\
&\frac{\partial c_{12|13|21|31}}{\partial \zeta} = -\frac{8}{9}\mu \\
&\frac{\partial c_{55}}{\partial \mu} = 1; \frac{\partial c_{32|23}}{\partial \mu} = -\frac{2}{3}\left(1 - \frac{2}{3}\zeta\right), \\
&\frac{\partial c_{32|23}}{\partial \zeta} = \frac{4}{9}\mu;
\end{aligned} \tag{4}$$

$\frac{\partial \sigma}{\partial \theta_i}$ is complex-valued in the time-harmonic regime, so the matrix 2-norm yields a real-valued stability estimate. We have spatially-resolved maps or estimates of θ and ε, so similar maps of the local stability estimate can be computed for each parameter. In a clinical setting, evaluating the TIPS metric immediately after data collection would use θ from an atlas, or the metric could be computed after TI-NLI inversion with the final subject-specific property maps in cases where large deviations from population averages are expected, in order to give a post hoc assessment of property estimate stability. The TIPS metric is calculated for each voxel across the entire image.

Stability analysis

We study the correlation between the computed stability measure, $\Delta_{\sigma,\theta_1}(\vec{x})$, and the observed noise sensitivity. For simulated data, the noise sensitivity was computed for each recovered parameter as the absolute value of the difference of an inversion with Gaussian noise of 2% or 5% added to the displacement data used in the inversion, and the un-noised (reference) inversion. The un-noised inversion served as the reference instead of the ground truth to avoid obscuring the noise sensitivity with the somewhat larger errors from incomplete contrast recovery due to the regularization required for *in vivo* stability. We calculate mean noise sensitivity as a function of stability by binning voxel-wise errors into equally-sized bins based on their TIPS value. Spearman coefficient and p-value were calculated to establish the monotonically decreasing relationship between these two quantities.

Experiment 1. Transversely isotropic spherical shell simulation: multiple actuations study

Simulated displacement data of a spherical shell centered inside a cube (figure 1) was analyzed. The outer cube was isotropic whereas the inner spherical shell was transversely isotropic with fibers running parallel to the surface of the shell, as shown. Cube length was 90 mm and the 10 mm thick shell had a diameter of 80 mm. We studied MRE reconstruction of this system by actuating (vibrating) one of the six faces of the cube and estimating the material properties from the displacement field. Other faces of the cube were stress-free.

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Realistic values for the shear modulus and damping ratio from isotropic brain MRE values in the literature [23] were selected. Values for tensile anisotropy (ζ) and shear anisotropy (φ) in brain tissue have not been evaluated comprehensively; however, properties in the range of 0–1 are expected [24–27]. Several values in this range were chosen to avoid the special case where $\phi = \zeta$. Mechanical properties assigned to the shell and background for each experiment are given in table 1. Synthetic data was generated by solving the forward problem for the displacement field at high resolution (1.5 mm) to reduce discretization error. Simulated finite resolution measurements were produced by interpolating to a typical MRE resolution of 2.05 mm. Our subzone-based TI-NLI inversion algorithm established in recent work [17] was used to reconstruct μ, ζ, and φ from the simulated displacements.

The cubic outer geometry in this simulation allowed us to control the propagation and polarization directions of the shear waves by applying fixed displacement boundary conditions to the X/Y/Z = 0 faces of the cube, such as X = 0-face, Z-direction (abbreviated as XZ). The configuration yielded nine displacement sets, 3 with motion normal to the face and six with shear motion. We present a broad summary of MRE inversions of these nine displacements, including some with multiple wave fields.

Experiment 2. Transversely isotropic spherical shell simulation: stability study

We focused on noise sensitivity with the computed parameter stability metric for two important actuations of the shell simulations, namely XX and ZZ. These cases are the counterparts of AP and LR actuations in brain MRE. The true property values and simulated displacements were used in the Δ_{σ,θ_i} calculations for simulated data.

Experiment 3. Transversely isotropic brain simulation based on DTI data: stability study

We used a previously published transversely isotropic MRE simulation of the human brain [22] with fiber directions defined from DTI data, realistic boundary conditions applied from an *in vivo* multi-direction MRE exam [26], and realistic values of NITI parameters assigned to gray matter, white matter, 6 subcortical gray matter regions, and 11 white matter tracts. Mechanical properties for the brain simulation are summarized in table 2. We assumed that the mechanical anisotropy (both ζ and ϕ) was approximately the same as the DTI anisotropy (figure 4), with a random multiplier between 0.8–1.2 applied to each anisotropy type to avoid the special case where $\phi = \zeta$.

Inversions from each of two separate displacement fields—from anterior-posterior (AP) and left-right (LR) actuation—were analyzed separately as well as together to evaluate the value of our sensitivity measure. A recent previous publication [17] on this model assumed mechanical anisotropy was 20% of DTI anisotropy. Here, we applied higher mechanical anisotropy (of approximately 100% of DTI anisotropy) since the published tensile anisotropy values fall in this range.

Experiment 4. *In vivo* TI-NLI: stability study

Finally, we considered TI-NLI MRE reconstruction of human brain *in vivo* displacement data from a single displacement field and performed the parameter stability calculation. We computed the pixel-wise mean and standard deviation of the material properties for a set of

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10 registered repeat MRE acquisitions from a single subject with a one displacement field (vibration in the AP direction).

The procedure to calculate the p-value for the correlation between computed stability and noise sensitivity was the same as described above, except that the absolute difference between the mean value of 10 registered repeated inversions and a single registered repeat was used to approximate the property variability map, which was applied instead of the noise sensitivity. We compared the computed parameter stability with the observed variability for the repeats with a representative low and high average $\Delta_{\sigma,\zeta}$. The mean TI mechanical property map over the 10 repeats, and the measured MRE displacements were used to compute Δ_{σ,θ_i} for *in vivo* data.

The mean over 10 repetitions was used to create the material property map when computing the stability map for each repeat, with the idea that a property atlas [3] would be used once more data is available. The TIPS score does change slightly if a different property map is used, e.g. from a single repeat, but the change is minor, about 5%. In other words, it only perturbs the data points in figure 9 and does not change the overall trend dramatically.

Results and discussion

Experiment 1. Shell simulation: multiple actuations

Figure 2 shows a representative horizontal slice of reconstructed mechanical properties for TI-NLI inversions of simulated data. Results are shown for inversions using data from one (e.g. XX), two (e.g. XX, ZZ), three (XX, YY, ZZ), and finally all nine actuations. In general, accuracy of reconstructions improved significantly with number of actuations, but with diminishing gains. Focusing on tensile anisotropy ζ, we observe that ZZ was better than XX. The combination of XX, ZZ actuations improved contrast significantly compared to either actuation alone. Including three actuations (XX, YY, ZZ) improved results more modestly relative to XX, ZZ. Normal actuations yielded significantly better ζ reconstructions in this case compared to shear actuations (XZ and so on) do. Likewise, shear actuations yielded better reconstructions of shear anisotropy, φ. Some ‘crosstalk’ was observed in the shear modulus μ (i.e. being overestimated in the shell region), but crosstalk was reduced when data from more actuation directions were included. Using all 9 actuations provided the best results, presumably due to the richness of the displacement data from the combined wave fields.

Experiment 2. Shell simulation: parameter stability calculation

Initial *in vivo* results suggest that ζ images have the most correspondence to anatomy [17]; therefore, we focused on ζ images in the following analysis, although similar analysis for ϕ yielded similar results. Figure 3 shows noise sensitivity and parameter stability of ζ reconstructions for a representative slice for XX and ZZ actuations, respectively. Significant anti-correlation ($p<0.01$) was found in the XX case, and the images showed that areas with low TIPS corresponded to regions where the inversion had low stability and high noise sensitivity.

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For the ZZ case, the parameter stability map was nearly homogeneous with reasonably high values and no statistically significant anti-correlation was evident, and all noise sensitivity values were lower than in the noisy region of the XZ case. Thus, our stability metric is able to predict areas of low versus high sensitivity to noisy data in the inverse problem.

Experiment 3. Brain simulation: parameter stability maps

Figure 4 focuses on a representative axial slice and shows the material property truth values (panels a, c and e) and their respective reconstructions (panels b, d and f).

Reconstructions have good fidelity to the truth. White matter regions can be distinguished from their gray matter counterparts (figure 5(a) has the anatomical MRI (T_2) used to generate the simulation for reference). As with the shell simulations, incomplete contrast recovery for ζ and φ of smaller structures occurred due to the regularization required to maintain stability. Apparent crosstalk between parameter estimates led to overestimated μ values in regions with anisotropy i.e. $\zeta > 0$, similarly to the crosstalk observed in the shell simulations.

Reconstructions have good accuracy. Anti-correlation between noise sensitivity and TIPS maps is apparent in all three orthogonal slices. Correlation is apparent in all three AP sections, as well as the LR case. In figure 6, noise sensitivity versus stability is shown for the full brain for the AP and LR synthetic data, which shows a clear threshold of the stability metric above which low noise sensitivity can be expected. High Spearman coefficients and p-values within the scientific standard of 0.05 are observed.

Experiment 4. *In vivo* data: inversions from a single displacement field

Finally, TI-NLI MRE reconstruction of human brain *in vivo* displacement data demonstrated the utility of the stability calculation. Figure 7 panels b through g present the pixel-wise mean and standard deviation of the material properties for a set of 10 registered repeat MRE acquisitions from a single subject, using a single displacement field (vibration in the AP direction). For comparison, we show the corresponding plots for LR only (panels s-x) and combined AP and LR displacement datasets (panels l-r). Material property variabilities are significantly larger for LR relative to AP panels, and combined AP and LR data yielded reconstructions of similar quality to AP data alone. The axial slice (panel 7a) is located near to the same region as shown for the brain simulations (figure 5(a)). A region of high ζ values corresponding to the corona radiata is evident in this slice; black contours marking this tract are drawn in all the panels.

Small regions of slightly negative values for ζ, and larger regions of negative values for φ, are apparent. Left-right symmetry is evident in all three material properties, which indicates our reconstruction appears to be working well overall even with only a single displacement field. Mean ζ (panel c) has good correspondence with white matter regions in the MR image (panel a). Amplitude of φ is significantly less than ζ. The former is about plus/minus 0.5 and the latter ranges from 0 to 1.5, with some small regions going slightly less than zero.

For μ, some noise occurred (large standard deviation, panel e) along a thin line between the hemispheres; likely due to the cerebral falx, which is a thin, stiff membrane that incurs

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significant modeling errors with the continuous, 2 mm resolution support of the unknown properties being estimated in the TI-NLI algorithm. Also shown is ζ for representative orthogonal sagittal and coronal slices (panels h-k), located in the brain where the simulation slices appear in figure 5. Good correspondence exists between the anatomical MRI and reconstructions; ζ is high in white matter regions, and small or zero in gray matter areas.

Figure 8 show TIPS calculations applied to *in vivo* scans from a single subject. Two representative repeat acquisitions are presented; one high stability repeat study from AP actuation, and a low stability repeat acquisition from LR actuation. Reconstruction of data with high TIPS $\Delta_{\sigma,\zeta}$ values (panel **b**) appears closer (almost identical) to the reference mean (panel **a**) than reconstruction of low TIPS data (panel **f**) relative to its reference mean (panel **e**). The respective variability maps highlight this observation, panels **c** and **g**. For the low-TIPS repeat, anti-correlation between error (panel **e**) and sensitivity (panel g) is visually apparent. For the high TIPS repeat, error and $\Delta_{\sigma,\zeta}$ maps are more homogeneous; a few areas appear with low $\Delta_{\sigma,\zeta}$. Hence, the variability map (panel **c**) also looks more homogeneous than its stability counterpart (panel **g**).

Finally, TIPS scores are shown in figure 9 for the different repeat acquisitions for two representative white matter tracts, namely the corona radiata and corpus callosum. Instead of using data from a single axial slice, every voxel inside the bounds of the three-dimensional tract was considered. Variability was computed as the mean values over all voxels of the voxel-wise absolute difference of each repeat with the mean of all ten repeats, and the TIPS score is the mean over all voxels of the stability metric. The repeat numbers are annotated to allow individual comparisons with AP, LR and APLR acquisitions.

The LR repeats have lower TIPS score on average and higher variability compared to the AP repeats. Variability decreases as TIPS increases; high stability implies low variability can be expected and the data can be trusted. The TIPS scores for APLR were estimated by taking a simple mean of the AP and LR scores; those data points fit the overall trend well. The Spearman coefficient R was calculated for all 30 data points, and indicate strong monotonically decreasing correlation with associated p-values well below the 0.05 scientific standard.

In figure 10, we show *in vivo* TI-NLI inversions from single AP displacement for 5 additional subjects spanning a range of TIPS scores. Although it is difficult to quantify the accuracy of the TI-NLI reconstructions from a single scan because reliable independent measurements of the true values are not available, qualitatively, the ζ images of the corona radiata (which is expected to have higher than average anisotropy due to a high density of bundled myelinated fibers) improve with higher TIPS score. This supports the findings from analysis of simulated data and repeated *in vivo* data and suggests TIPS will be a useful metric for evaluating the quality of MRE data for anisotropic MRE applications.

Conclusion

Analysis of data from simulated MRE in an anisotropic shell phantom demonstrated that accuracy of TI-NLI reconstructions improve with more actuation directions, but with

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diminishing gains. For the idealized shell simulation data, reconstruction recovered the shape of the spherical shell with few artifacts with data from just two actuations, i.e. XX and ZZ. RMS error relative to ground truth improved by about 13%, if data from all nine possible actuations were used in the ζ reconstruction compared to just two (XX and ZZ). For brain simulations, reconstructions recovered μ, ζ and φ of white matter tracts with a single actuation (AP), which would reduce imaging time, and can use the same experimental vibrations systems available for traditional brain MRE, rather than multi-direction actuators [28]. RMS error improved by about 2% when both datasets AP + LR were used instead of either LR or AP alone. For *in vivo* acquisition data, variability and property estimates with AP were similar to two datasets, AP + LR. Lack of ground truth results makes *in vivo* data more difficult, if not impossible, to confirm the accuracy of inversion. Phantoms with known transversely-isotropic properties will be able to determine if more than one data set will be required experimentally for accurate transversely-isotropic property recovery.

The transversely isotropic parameter stability (TIPS) maps predict areas in each TI-NLI parameter image which have high sensitivity to measurement noise; these regions parameter estimates are expected to be less reliable. TIPS is a generalization of the widely used octahedral strain-SNR that has proved valuable for isotropic MRE. The utility of TIPS was established first by using data from simulations of an anisotropic shell and brain, and then with *in vivo* data. TIPS provides a simple metric that can be used to identify high- and low-quality data in future clinical applications of TI-NLI.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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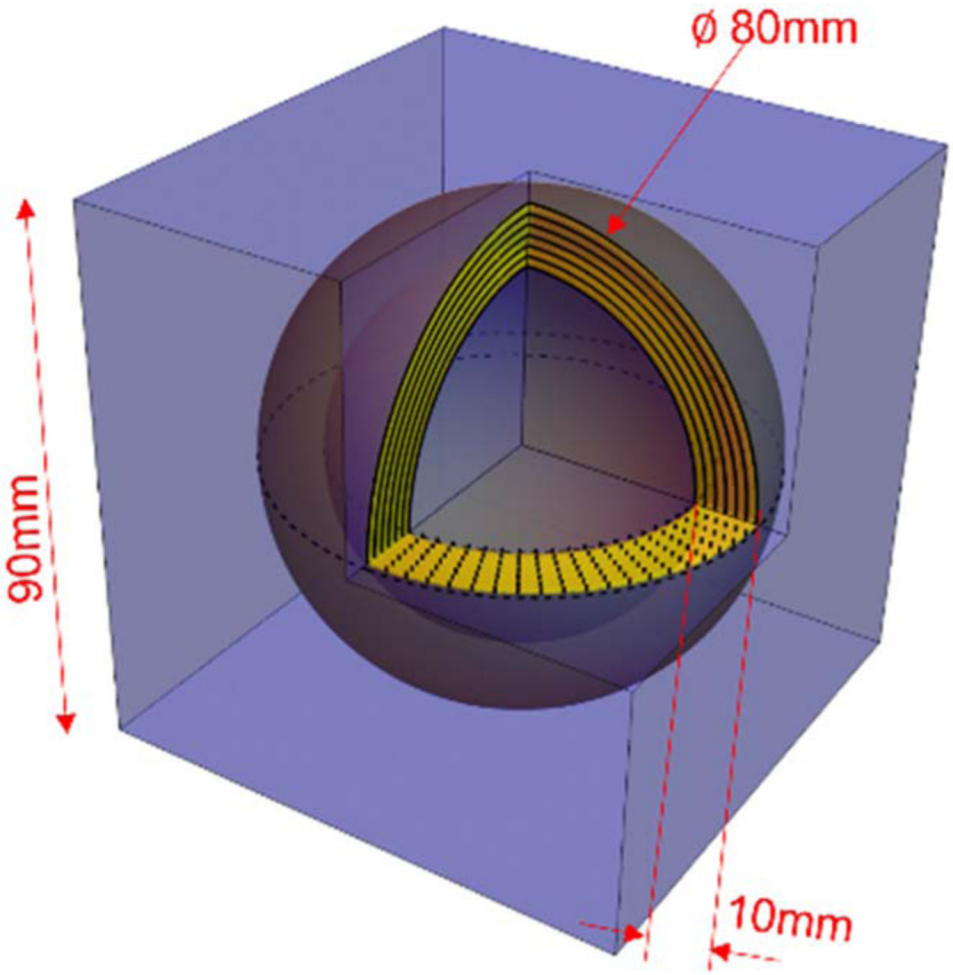

Figure 1.
Transversely isotropic spherical shell inside an isotropic cube.

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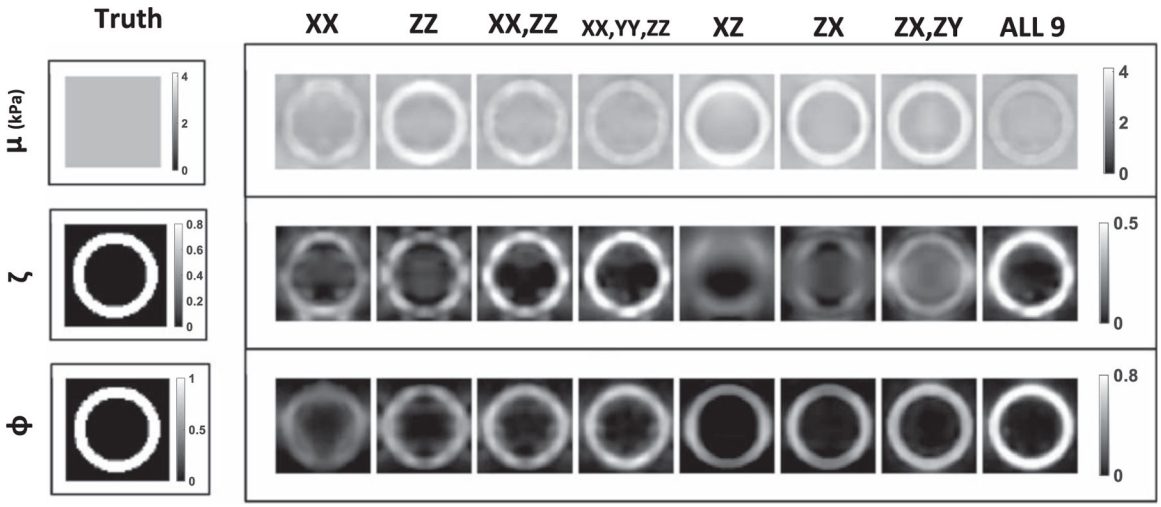

Figure 2.

Simulation MRE reconstruction of a transversely anisotropic spherical shell (figure 1) using combinations of actuations. Reconstruction generally improved with number of actuations, but with diminishing gains. Two normal actuations (e.g. XX, ZZ) were sufficient for producing a quality reconstruction.

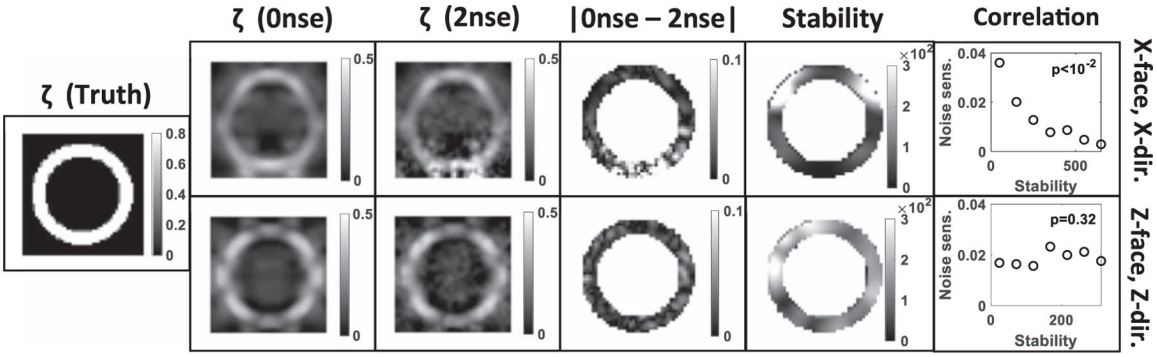

Figure 3.

Stability maps for a shell (figure 1). ζ reconstructions from two actuations, XX (top row) and ZZ (bottom row).

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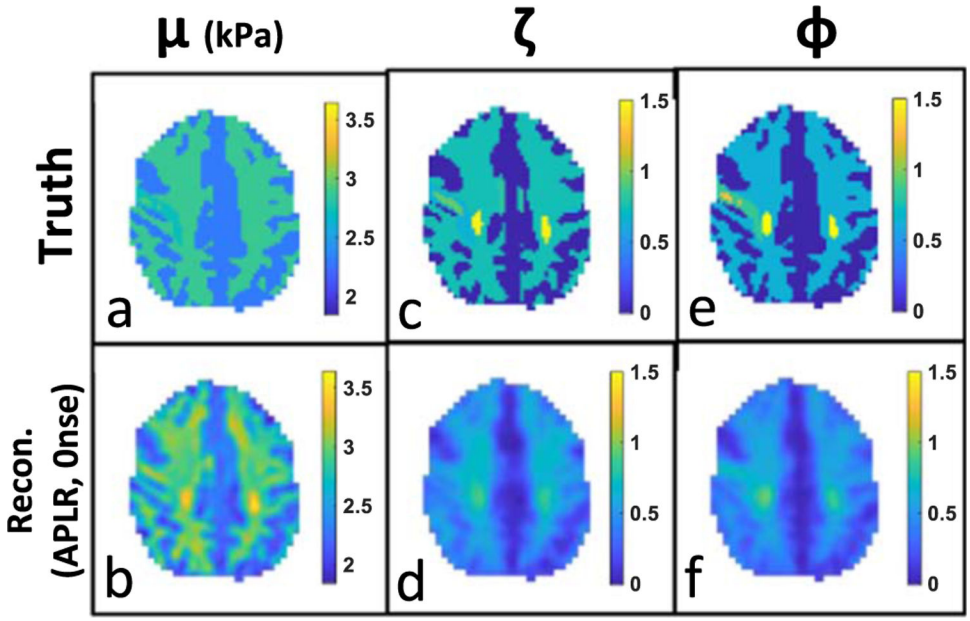

Figure 4.

Brain Simulation. Ground truth values of the three important parameters, and their respective reconstructions are shown in a single representative axial slice.

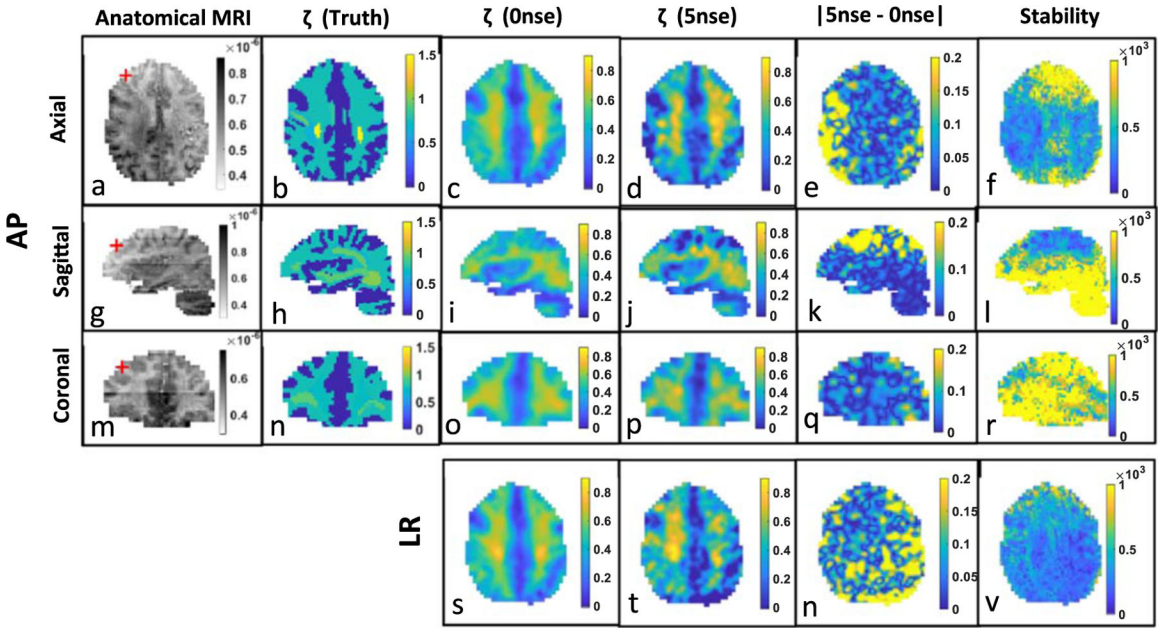

Figure 5.

Brain simulation TI-NLI recovered property images and stability maps for ζ. Single representative slices for three orthogonal planes that intersect at the red plus (+), see leftmost column. The anti-correlation of stability with noise sensitivity is apparent.

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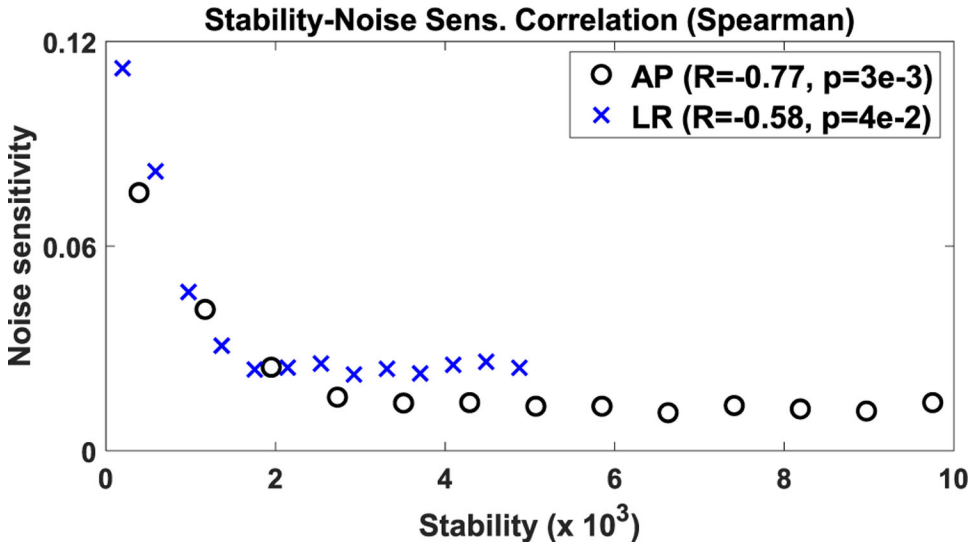

Figure 6.
Correlation between stability and noise sensitivity for the brain simulation.

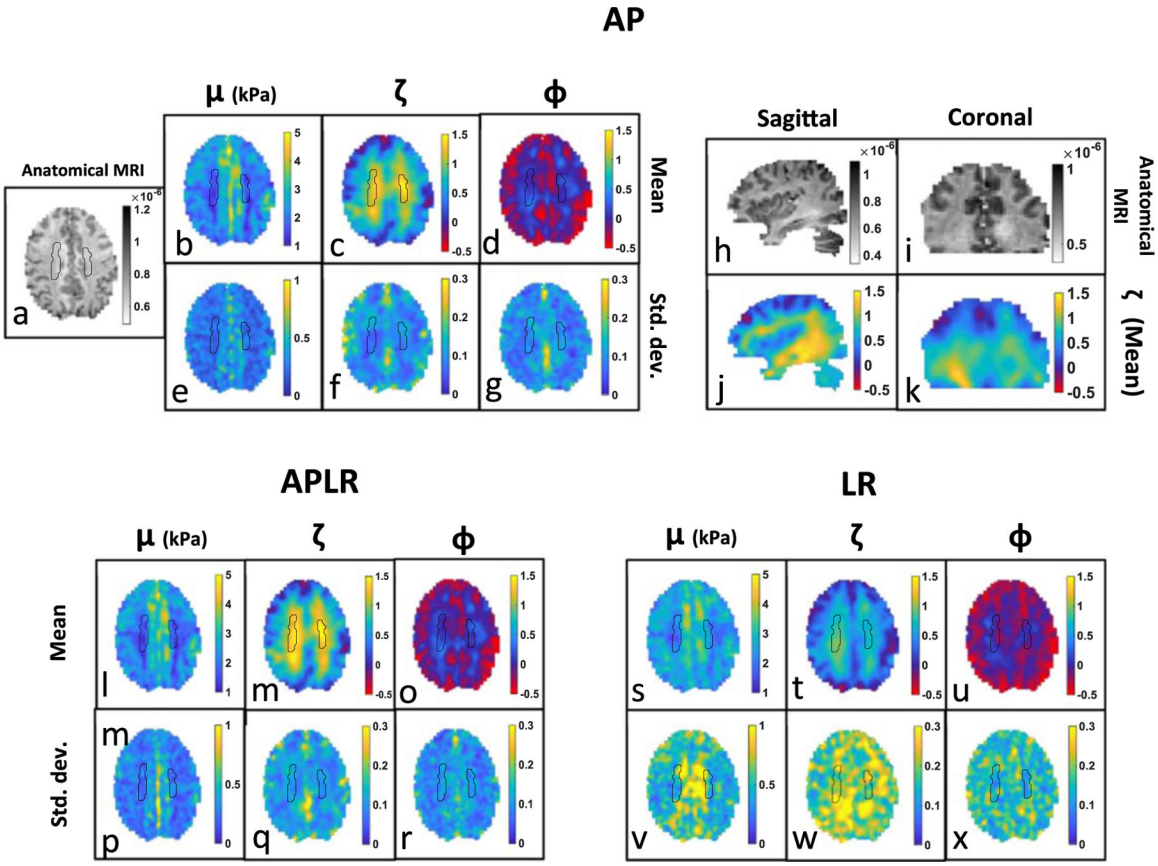

Figure 7.

Brain *in vivo* data (10 registered repeats from a single subject). TI-NLI inversions from displacement fields with AP and LR directed vibration fields, and AP + LR combined are show. The black contours are the left and right corona radiata.

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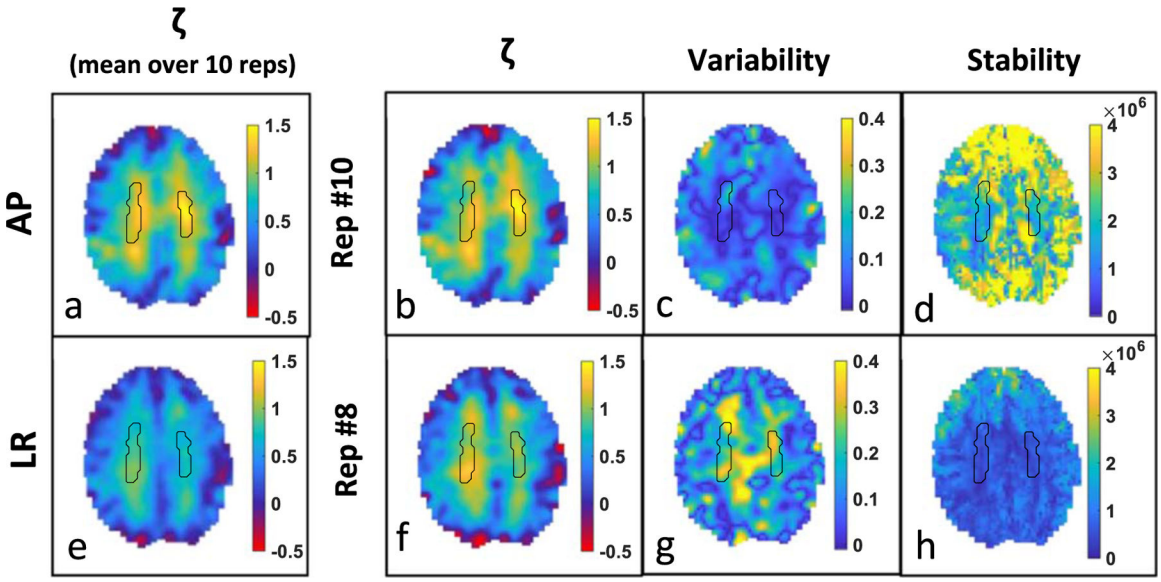

Figure 8.

TI-NLI inversions and brain *in vivo* stability maps for ζ for two of the repeated in vivo brain scans, a high stability (rep 10), and low stability (rep 8) example is shown. Black contours are the left and right corona radiata.

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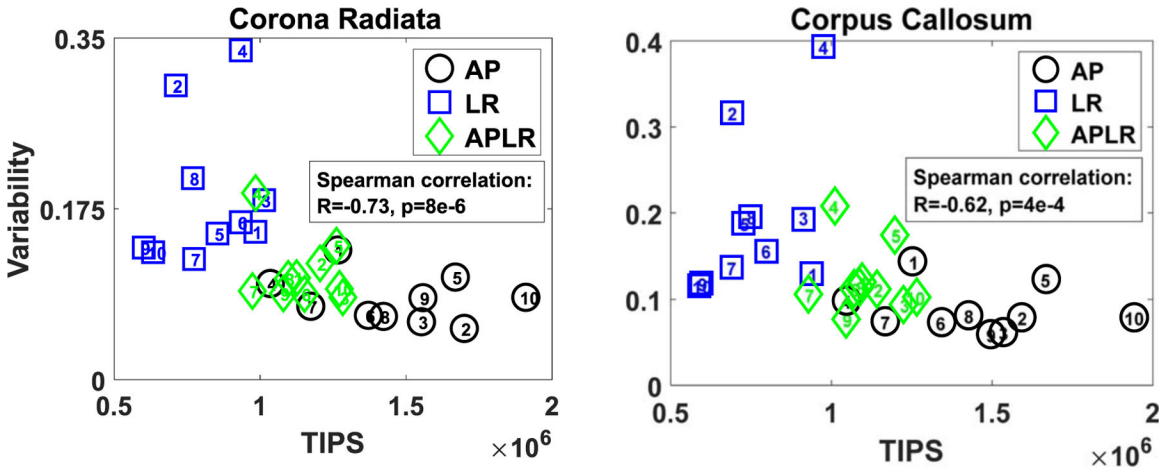

Figure 9.

In vivo brain TIPS scores for specific white matter tracts.

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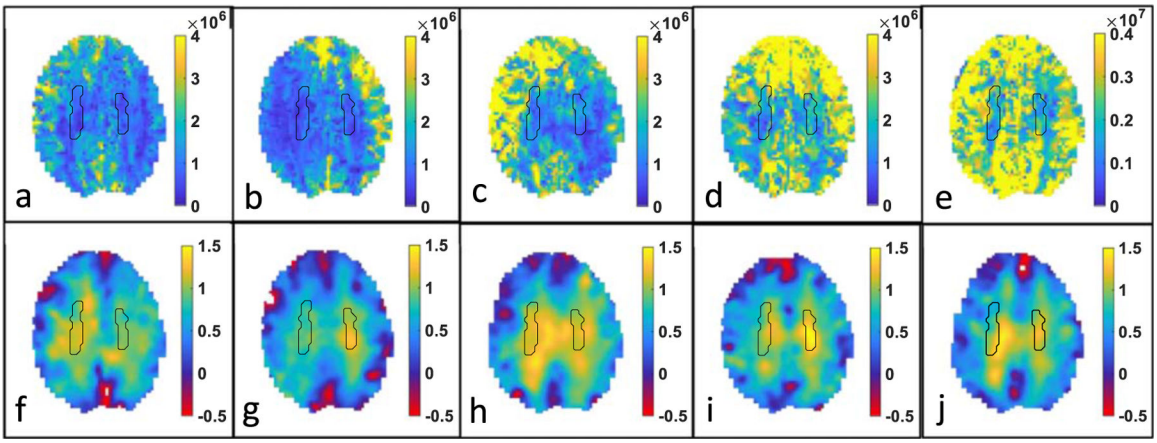

Figure 10.

Parameter Stability maps (top row), and Tensile anisotropy, ζ, (bottom row) computed with TI-NLI for *in vivo* brain MRE exams of 5 different subjects. A single 50 Hz displacement field with vibration in the anterior-posterior direction was used, Black contours are the left and right corona radiata.

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Table 1.

Simulation parameters for a transversely isotropic spherical shell (figure 1). Literature values from isotropic brain MRE were used.

Shear modulus, μ	Damping ratio, ξ	Tensile anisotropy, ζ	Shear anisotropy, ϕ
2.9 kPa	0.207	0.8	1

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Table 2.

Simulation parameters for the DTI-based brain mechanical anisotropy model. Realistic values from isotropic brain MRE were used for μ and ξ, and DTI FA values were used to estimate reasonable values for ϕ and ζ.

Shear modulus, μ	Damping ratio, ξ	Tensile anisotropy, ζ	Shear anisotropy, ϕ
2.9 kPa	0.207	0.613–1.461	0.643–1.72